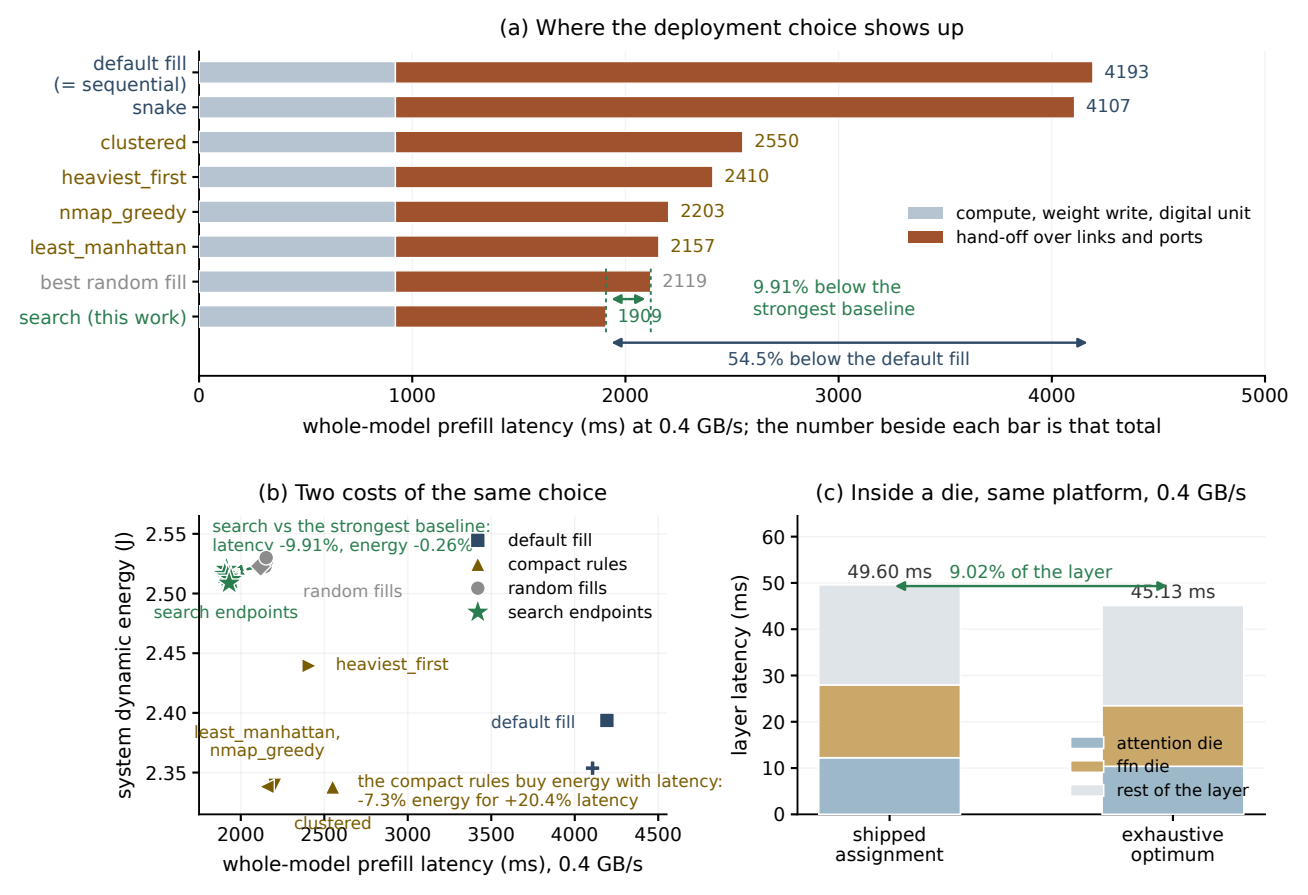
(a) Where the deployment choice shows up
default fill (= sequential)
snake
clustered
heaviest_first
nmap_greedy
least_manhattan
best random fill
search (this work)
4193
4107
2550
2410
2203
2157
2119
1909
compute, weight write, digital unit
hand-off over links and ports
9.91% below the strongest baseline
54.5% below the default fill
0
1000
2000
3000
4000
5000
whole-model prefill latency (ms) at 0.4 GB/s; the number beside each bar is that total
(b) Two costs of the same choice
search vs the strongest baseline:
latency -9.91%, energy -0.26%
random fills
search endpoints
heaviest_first
least_manhattan,
nmap_greedy
default fill
clustered
the compact rules buy energy with latency:
-7.3% energy for +20.4% latency
default fill
compact rules
random fills
search endpoints
system dynamic energy (J)
2.55
2.50
2.45
2.40
2.35
2000
2500
3000
3500
4000
4500
whole-model prefill latency (ms), 0.4 GB/s
(c) Inside a die, same platform, 0.4 GB/s
49.60 ms
9.02% of the layer
45.13 ms
attention die
ffn die
rest of the layer
layer latency (ms)
60
50
40
30
20
10
0
shipped assignment
exhaustive optimum